Improving Scottish Gaelic Machine Translation with NLLB, LLM Post-Editing, and Fine-Tuning

Danya Leyva, California State University Dominguez Hills
Benyamin Ahmadnia/ California State University Dominguez Hills
<dleyva14@toromail.csudh>; <bahmadniayebosari@csudh.edu>

Abstract

This project explores methods to improve Scottish Gaelic ↔ English machine translation in a low-resource setting. A baseline NLLB model was compared with LLM post-editing and fine-tuning on cleaned data. Systems were evaluated using BLEU, chrF, and COMET on a fixed test set. Results show that post-editing gave small improvements, while fine-tuning produced substantially larger gains, highlighting the importance of data quality and model adaptation.

Introduction

Neural machine translation performs well for high-resource languages but struggles for low-resource languages such as Scottish Gaelic due to limited and noisy training data. This project asks whether translation quality can be improved using LLM-based post-editing and fine-tuning. We compare baseline NLLB outputs, post-edited translations, and fine-tuned models to determine which approach provides the most effective improvements.

Methodology / Approach

- Selected Scottish Gaelic ↔ English
- Built baseline NLLB model
- Cleaned and filtered parallel data
- Applied LLM-based post-editing
- Fine-tuned on cleaned data
- Evaluated with BLEU, chrF, COMET

Background

Neural Machine Translation (NMT) models rely heavily on large, high-quality parallel datasets. Low-resource languages often lack sufficient clean data, leading to poor translation quality. Large Language Models (LLMs) can improve outputs through post-editing, but their effectiveness is unclear compared to directly improving training data. This study explores these trade-offs.

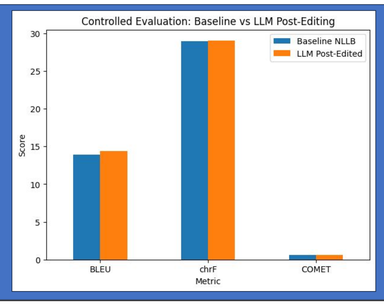
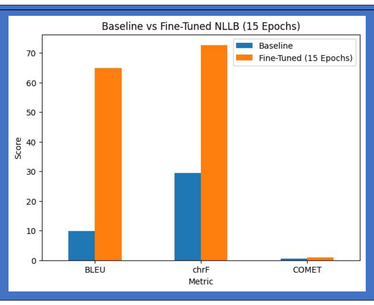


Figure 1: Baseline vs LLM Post-Editing across BLEU, chrF, and COMET.

Figure 2: Baseline vs Fine-Tuned NLLB (15 epochs) showing large improvements across BLEU, chrF, and COMET.



Results / Discussion

Results showed that LLM post-editing produced only modest improvements over the baseline model. In contrast, fine-tuning on cleaned data resulted in much larger gains across BLEU, chrF, and COMET. These findings suggest that while post-editing can provide small corrections, data quality and task-specific adaptation have a much larger impact on translation quality for low-resource settings. Fine-tuning consistently outperformed LLM post-editing across all metrics.

Future Work

- Improve dataset quality and filtering methods
- Evaluate on larger and more diverse test sets
- Explore additional low-resource language pairs
- Investigate methods to reduce overfitting during fine-tuning
- Compare different LLMs for post-editing effectiveness

Summary / Conclusion

This project demonstrates that improving data quality and fine-tuning the model are more effective than post-editing alone for Scottish Gaelic machine translation. The results highlight the importance of reproducible evaluation, fixed test subsets, and careful dataset cleaning in low-resource NMT research.

References

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Wolf et al., 2020. Transformers

Acknowledgements

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